Care Access Navigator

Incident Response 1 — Task 3: Incident Response and Containment

Network isolation, evidence preservation and a tested containment playbook, started by a live Wazuh alert (IR1-2026-002)

Document ID	CAN-IR1-T3-CONT-002 (replaces 001: same lab, people and incident numbers as Tasks 1–2 and 4–6)
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Date	November 3, 2026
Incident	**IR1-2026-002** SEV-1: Tara Jones (`test.nurse`, mover Nurse → Billing) read synthetic record PT-1003 through her old clinical group and sent it off the host. Also contained in the same session: IR1-2026-001 (leaver key) and IR1-2026-003 (sudo grant)
Trigger	Wazuh rule 100221 (level 12) on agent parrot-vm at 19:31:10.008 UTC; containment began at that alert
Host	parrot-vm (Parrot OS stand-in, QEMU VM built in Task 1) with Wazuh agent 001
Run	19:31:10–19:35:47 UTC, 2026-11-03; 29 evidence files hashed

Summary

Tara Jones read PT-1003 at 19:31:08 UTC, staged a copy, and started an uploader with a cron job to restart it. The Wazuh agent shipped the EHR audit line and rule 100221 fired 1.3 s later, at 19:31:10 UTC. That alert started this task. Containment followed playbook CAN-IR-PB-02 RB-02: a live snapshot, volatile evidence, then an egress block **49 s after the alert**. Nothing reached the drop site after the block (0 uploads after it; 10 in total). Memory, a full RAM image and the file artifacts were preserved with matching hashes at both ends. The host was then isolated at the hypervisor: clinical cable out, quarantine adapter in. The Wazuh agent was moved to a single forwarded port, so monitoring continued. Finally the process, persistence, the EHR portal and the account were shut down, and the two queued incidents were contained on the same host. The six probes from Task 1 were re-run: every production path went **REACHABLE** → **BLOCKED**, and only the forensic path opened.

1. How this report meets the Task 3 requirements

Requirement	Where	Evidence
Network isolation: interface configuration	§4.1	transcripts/07
Network isolation: basic firewall rules	§4.2	containment/iptables.containment.rules, kern.log
Network isolation: segmentation using (VirtualBox) networking	§4.3	transcripts/07, 09; Task 1 baseline
Evidence preservation: network traffic captures	§5.1	network/*.pcap

Requirement	Where	Evidence
Evidence preservation: basic memory dumps	§5.2	memory/, transcripts/05
Evidence preservation: file system artifacts	§5.3	filesystem/
Containment playbook: host isolation steps in VirtualBox	§3 (C1, C6)	VBoxManage column
Containment playbook: network traffic blocking	§3 (C3, C6d)	transcripts/04, 07
Containment playbook: service shutdown procedures	§3 (C7), §6	transcripts/08
Documentation and evidence of functionality	§7	timeline, measures, evidence index

2. From alert to decision

The alert that started containment (task2 evidence/wazuh/alerts.json)


```
{
 "timestamp": "2026-11-03T19:31:10.008+0000",
 "agent": {
  "id": "001",
  "name": "parrot-vm",
  "ip": "10.0.2.15"
 },
 "location": "/var/log/ehr-audit.log",
 "full_log": "2026-11-03T19:31:08.718332+00:00 parrot-vm ehr-audit: user=test.nurse action=VIEW
    patient=PT-1003 decision=ALLOW basis=group:clinical",
 "rule": {
  "id": "100221",
  "level": 12,
  "description": "ePHI viewed by mover test.nurse (now Billing) through stale clinical group - record
    PT-1003"
 }
}
```

transcripts/01_triage.txt - triage on the host

```
$ tail -2 /var/log/ehr-audit.log
2026-11-03T19:29:51.802321+00:00 parrot-vm ehr-audit: user=test.clinician action=VIEW patient=PT-1001
    decision=ALLOW b
2026-11-03T19:31:08.718332+00:00 parrot-vm ehr-audit: user=test.nurse action=VIEW patient=PT-1003
    decision=ALLOW basis

$ ausearch -k ehr-read -i | grep -m2 PT-1003
type=PROCTITLE msg=audit(11/03/26 19:31:08.733:470) : proctitle=cat /srv/ehr/clinical/PT-1003.txt
type=PATH msg=audit(11/03/26 19:31:08.733:470) : item=0 name=/srv/ehr/clinical/PT-1003.txt inode=31804
    dev=fe:01 mode=

$ timeout 12 tcpdump -nn -i enp0s1 -c 3 'dst host 10.78.99.66 and tcp[tcpflags] & tcp-syn != 0'
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
19:31:15.587955 IP 10.0.2.15.55034 > 10.78.99.66.8080: Flags [S], seq 601044798, win 64240, options [mss
     1460,sackOK,T
19:31:20.602020 IP 10.0.2.15.51488 > 10.78.99.66.8080: Flags [S], seq 1960232890, win 64240, options
    [mss 1460,sackOK,

2 packets captured
3 packets received by filter
0 packets dropped by kernel

$ ps -o pid,user,lstart,cmd -u test.nurse
    PID USER                     STARTED CMD
   1474 test.nu+ Tue Nov  3 19:31:06 2026 /lib/systemd/systemd --user
   1475 test.nu+ Tue Nov  3 19:31:06 2026 (sd-pam)
   1534 test.nu+ Tue Nov  3 19:31:09 2026 python3 /home/test.nurse/.cache/.sync/syncd.py

# Finding: PT-1003 is being sent off-host to 10.78.99.66:8080 by a process owned by test.nurse.
# IR1-2026-002 confirmed SEV-1 (Critical): ePHI disclosure in progress -> containment authorised.
# Queue: IR1-2026-003 (SEV-2, unapproved sudo) and IR1-2026-001 (SEV-3, leaver attempts) are contained
    on the same hos
```

3. Containment playbook CAN-IR-PB-02 RB-02 (executed as written)

The order follows volatility: capture what disappears first, stop the loss as soon as it is captured, and keep every change reversible until the evidence is safe.

Step	Action	Executed here	VirtualBox / Parrot OS
C1	Live snapshot (undo point; preserves RAM + disk)	HMP savevm ir-pre-containment	VBoxManage snapshot parrot-vm take ir-pre-containment --live
C2	Volatile data: packets, processes, connections, persistence	tcpdump 25 s; ps/ss/lsof/proc over the guest agent	same tools; output to case storage
C3	**Block the traffic**, keep the process alive	iptables chain IR-EGRESS-BLOCK on OUTPUT	same iptables rules
C4	Memory: process core + full RAM	gdb gcore; QMP dump-guest-memory	gcore; VBoxManage debugvm dumpvmcore
C5	File artifacts: stat → freeze → hash → tar → verify	stat, tar to RAM, sha256sum, tar over SSH	same
C6	**Isolate**: cable out, quarantine adapter in, default-deny, keep the SIEM	QMP set_link; ip; iptables -P DROP; agent via 10.0.3.20:1514	setlinkstate1 off; setlinkstate2 on (intnet quarantine)
C7	**Shut down** process, persistence, exposed service; lock account; contain the queued incidents	pkill; mask cron; disable portal; usermod -L; gpasswd -d; keys moved	same systemctl / usermod
C8	Verify with the Task 1 probes; snapshot	probe matrix; savevm ir-contained	VBoxManage snapshot ... take ir-contained

Design choice: The uploader (PID 1534) was not killed when found. Killing it would have lost its memory, which holds the record and the drop-site URL (§5.2). It was cut off from the network first and killed only after the memory images were hashed.

4. Network isolation

4.1 Interface configuration

transcripts/07 - hypervisor link change and guest interfaces

```
# C6a pull the clinical cable   [setlinkstate1 off] ; C6b quarantine adapter in   [setlinkstate2 on]
{"return": {}}
{"return": {}}
nic0: index=0,type=nic,model=virtio-net-pci,macaddr=52:54:00:0a:00:10
 \ n0: index=0,type=user,net=10.0.2.0,restrict=off
  - cap0: type=filter-dump,file=/vm/nic0-clinical.pcap,maxlen=65536,position=tail,status=on,insert=behin
    d,netdev=n0,qu
nic1: index=0,type=nic,model=virtio-net-pci,macaddr=52:54:00:0a:00:50
 \ q0: index=0,type=user,net=10.0.3.0,restrict=on
  - cap1: type=filter-dump,file=/vm/nic1-quarantine.pcap,maxlen=65536,position=tail,status=on,insert=beh
    ind,netdev=q0,

# C6c guest interface configuration
lo               UNKNOWN        00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
enp0s1           DOWN           52:54:00:0a:00:10 <BROADCAST,MULTICAST>
enp0s2           DOWN           52:54:00:0a:00:50 <NO-CARRIER,BROADCAST,MULTICAST,UP,LOWER_UP>
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s1           DOWN
enp0s2           DOWN           10.0.3.15/24
default via 10.0.3.2 dev enp0s2
10.0.3.0/24 dev enp0s2 proto kernel scope link src 10.0.3.15
```

4.2 Firewall rules

Two stages. **C3** drops and logs everything to the IOC while leaving the host up for evidence collection. **C6d** sets every policy to DROP and opens two paths only: SSH from the forensic workstation on the quarantine interface, and the Wazuh agent to the quarantine SIEM forward. Before the incident the ruleset was empty (`iptables.pre-containment.rules`): the host had no firewall, a finding for Task 6.

containment/iptables.containment.rules

```
# Generated by iptables-save v1.8.9 (nf_tables) on Tue Nov  3 19:32:47 2026
*filter
:INPUT DROP [0:0]
:FORWARD DROP [0:0]
:OUTPUT DROP [2:95]
:IR-EGRESS-BLOCK - [0:0]
-A INPUT -i lo -j ACCEPT
-A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A INPUT -s 10.78.50.5/32 -i enp0s2 -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -m limit --limit 20/min -j LOG --log-prefix "IR-DENY-IN IR1-2026-002 "
-A OUTPUT -d 10.78.99.66/32 -j IR-EGRESS-BLOCK
-A OUTPUT -o lo -j ACCEPT
-A OUTPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A OUTPUT -d 10.0.3.20/32 -o enp0s2 -p tcp -m tcp --dport 1514 -j ACCEPT
-A IR-EGRESS-BLOCK -m limit --limit 20/min -j LOG --log-prefix "IR-BLOCK IR1-2026-002 " --log-uid
-A IR-EGRESS-BLOCK -j DROP
COMMIT
# Completed on Tue Nov  3 19:32:47 2026
```

The kernel logged 15 IR-BLOCK lines before the artifacts were frozen, each tagged `UID=1003` (test.nurse) by `--log-uid`. The firewall itself names the account behind every blocked packet.

4.3 Segmentation and monitoring: before and after

#	Path	Before (Task 1)	After	Result
P1	macOS host -> parrot-vm SSH (127.0.0.1:2223)	**REACHABLE**	**BLOCKED**	**PASS**

#	Path	Before (Task 1)	After	Result
P2	clinical LAN nurse-ws -> parrot-vm SSH (10.78.10.5:22)	**REACHABLE**	**BLOCKED**	**PASS**
P3	clinical LAN nurse-ws -> EHR portal (10.78.10.5:8080)	**REACHABLE**	**BLOCKED**	**PASS**
P4	parrot-vm -> exfil-sink (10.78.99.66:8080)	**REACHABLE**	**BLOCKED**	**PASS**
P5	parrot-vm -> internet (deb.debian.org:80)	**REACHABLE (HTTP 200)**	**BLOCKED**	**PASS**
P6	ir-forensics -> parrot-vm quarantine SSH (10.78.50.2:2222)	**BLOCKED**	**REACHABLE**	**PASS**

SIEM kept during quarantine. The quarantine network has no route anywhere. Its only exit is one forward the hypervisor provides, 10.0.3.20:1514 → wazuh-manager. The agent was re-pointed to it and reconnected at 19:34:02 UTC, so containment and everything after it stayed visible to the SIEM.

transcripts/07 (C6e)

```
# C6e keep monitoring: the Wazuh agent reports through the hypervisor's single forward 10.0.3.20:1514 ->
     wazuh-manager
2026/11/03 19:34:02 wazuh-agentd: INFO: (4102): Connected to the server ([10.0.3.20]:1514/tcp).
   Status:     Active
   Last keep alive:     1793734449
```

5. Evidence preservation with Parrot OS tools

5.1 Network traffic captures

Capture	Taken by	Packets	Purpose
`IR1-2026-002_live_tcpdump.pcap`	tcpdump on parrot-vm (C2a)	60	Host-side proof of the uploads
`hypervisor_nic0_clinical.pcap`	hypervisor NIC trace (=--nictrace)	22 k	All adapter-1 traffic for the whole exercise; the guest cannot tamper with it
`hypervisor_nic1_quarantine.pcap`	hypervisor NIC trace	1756	Only forensic SSH and the agent after isolation

The hypervisor capture holds **10** HTTP POSTs to the IOC, the same number the drop site logged (10). The live capture's hash matches at both ends (MATCH).

5.2 Memory dumps

Dump	Tool	Size	SHA-256	Stored
Process core of PID 1534 (syncd.py)	gdb gcore (guest)	12.0 MB	`1d4ccb577d0dcdd5b1cd…` **MATCH**	~/ir1-vm-lab/evidence (outside git)
Full RAM of parrot-vm	QMP dump-guest-memory	1.00 GiB	`7010d346a57e2818d14b…`	~/ir1-vm-lab/evidence (outside git)

memory/syncd_core_strings.txt - what the uploader held in memory

```
      7 /home/test.nurse/.cache/.sync/syncd.py
      3 10.78.99.66
      2 MRN: SYN-0003
      2 10.78.99.66:8080
      1 ~/.cache/.sync/pt1003.txt
      1 unclosed file <_io.BufferedReader name='/home/test.nurse/.cache/.sync/pt1003.txt'>
      1 syncd/0.3
      1 syncd.py
      1 pt1003.txt
      1 http://10.78.99.66:8080/sync
      1 er-Agent: syncd/0.3
      1 User-Agent: syncd/0.3
```

5.3 File-system artifacts

Metadata was read with stat *before* anything opened the files. Live logs keep growing, so a point-in-time copy was frozen into RAM, hashed, sent as a tar over SSH, and every file in the received tar was re-hashed.

Artifact	Owner	Mode	Modified (UTC)	Match
`/home/test.nurse/.cache/.sync/pt1003.txt`	test.nurse:test.nurse	644	2026-11-03 19:31:08	**MATCH**
`/home/test.nurse/.cache/.sync/syncd.py`	test.nurse:test.nurse	700	2026-11-03 19:31:09	**MATCH**

Artifact	Owner	Mode	Modified (UTC)	Match
/var/spool/cron/crontabs/test.nurse	test.nurse:crontab	600	2026-11-03 19:31:11	MATCH
/home/test.nurse/.ssh/authorized_keys	test.nurse:test.nurse	600	2026-11-03 19:21:04	MATCH
/home/test.billing/.ssh/authorized_keys	test.billing:test.billing	600	2026-11-03 19:21:04	MATCH
/var/log/auth.log	root:adm	640	2026-11-03 19:32:42	MATCH
/var/log/ehr-audit.log	root:adm	640	2026-11-03 19:31:08	MATCH
/var/log/kern.log	root:adm	640	2026-11-03 19:32:39	MATCH
/var/log/audit/audit.log	root:adm	640	2026-11-03 19:32:42	MATCH
/etc/group	root:root	644	2026-11-03 19:30:03	MATCH
/etc/passwd	root:root	644	2026-11-03 19:21:04	MATCH
/etc/sudoers.d/it_admin	root:root	440	2026-11-03 19:21:03	MATCH
/var/ossec/logs/ossec.log	root:wazuh	660	2026-11-03 19:23:10	MATCH

6. Service shutdown and account containment (all three incidents)

transcripts/08_c7_shutdown.txt

```
## IR1-2026-002 (Tara Jones, mover)
$ pkill -KILL -f "^python3 .*[s]yncd"; pkill -KILL -u test.nurse; sleep 1; pgrep -u test.nurse || echo
    no test.nurse p
no test.nurse processes
$ systemctl stop cron && systemctl mask cron && systemctl is-active cron
Created symlink /etc/systemd/system/cron.service → /dev/null.
inactive
$ systemctl stop ehr-portal; systemctl disable ehr-portal 2>&1; echo "active=$(systemctl is-active
    ehr-portal) enabled
Removed "/etc/systemd/system/multi-user.target.wants/ehr-portal.service".
active=inactive enabled=disabled
$ usermod -L -e 1 test.nurse && gpasswd -d test.nurse clinical && usermod -aG disabled-users test.nurse
    && id test.nur
Removing user test.nurse from group clinical
uid=1003(test.nurse) gid=1006(test.nurse) groups=1006(test.nurse),1002(billing),1003(disabled-users)
$ install -d -m 700 /root/ir-quarantine && mv /home/test.nurse/.ssh/authorized_keys
    /root/ir-quarantine/test.nurse.aut
test.nurse.authorized_keys

## IR1-2026-003 (Sathya Hasi, unapproved sudo grant)
$ gpasswd -d test.clinician sudo && getent group sudo
Removing user test.clinician from group sudo
sudo:x:27:it_admin

## IR1-2026-001 (Jeevan Smith, leaver: blocked by DenyGroups, but the SSH key was still on the host)
$ mv /home/test.billing/.ssh/authorized_keys /root/ir-quarantine/test.billing.authorized_keys && ls
    /root/ir-quarantin
test.billing.authorized_keys
test.nurse.authorized_keys

## sshd: listen on the quarantine address only, responder only
$ printf "ListenAddress 10.0.3.15\nAllowUsers ir-analyst\n" >
    /etc/ssh/sshd_config.d/00-ir-containment.conf && sshd -t
listenaddress 10.0.3.15:22
allowusers ir-analyst
$ su -s /bin/sh test.nurse -c true 2>&1; echo "login as test.nurse: exit=$?"; setpriv --reuid=test.nurse
     --regid=test.
kex_exchange_identification: read: Connection reset by peer
```

Target	Procedure	Why
Uploader + test.nurse processes	pkill after memory was preserved	Ends the transfer; evidence already hashed
cron (persistence)	stop + mask	The */5 job would restart the uploader; the crontab is kept as evidence
ehr-portal	stop + disable	A compromised host must not serve ePHI; restored in Task 6
test.nurse (IR1-2026-002)	lock + expire; remove clinical; keys moved	Removes the root cause on this host
test.clinician sudo (IR1-2026-003)	gpasswd -d test.clinician sudo	The grant had no change record
test.billing key (IR1-2026-001)	authorized_keys moved to /root/ir-quarantine	The leaver's standing credential
sshd	ListenAddress quarantine only; AllowUsers ir-analyst	Kept as the forensic channel for Task 4

7. Containment timeline and measures

UTC	Phase	Event
19:31:10	detection	Wazuh rule 100221 alert (SEV-1)
19:31:12	triage	Host triage confirms disclosure in progress
19:31:26	c1_snapshot	C1 live snapshot
19:31:28	c2_volatile	C2 volatile evidence
19:31:59	c3_block	C3 egress blocked
19:32:32	c4_memory	C4 memory preserved
19:32:47	c5_filesystem	C5 artifacts preserved
19:32:52	c6_isolate	C6 isolated, SIEM kept
19:34:12	c7_shutdown	C7 shutdown; IR1-2026-001 and -003 contained
19:34:28	c8_verify	C8 probes re-run
19:35:47	contained	Snapshot ir-contained

Measure	Value
Record read → Wazuh alert	1.3 s
Alert → egress blocked	**49 s**
Uploads: total / after the alert / after the block	10 / 10 / **0**
Alert → isolated	102 s
Alert → contained and verified	277 s

8. Limitations and hand-off

- VirtualBox → QEMU and Parrot OS → Debian 12; each step lists its VirtualBox command.
- Eradication and recovery are left to Task 6, and the forensic analysis to Task 4.
- The lab stays running in the contained state, with snapshots baseline-clean, ir-pre-containment and ir-contained.

Appendix A — Evidence index

evidence/EVIDENCE-INDEX-SHA256.txt

```
7354c2f8754dab56...  ./containment/iptables.containment.rules
e3b0c44298fc1c14...  ./containment/iptables.pre-containment.rules
9b60c46749c9e3ee...  ./exfil-sink/uploads.jsonl
08f2adc1890a41a8...  ./filesystem/artifact_sha256_copy.txt
08f2adc1890a41a8...  ./filesystem/artifact_sha256_source.txt
10a9ec6df36dd2df...  ./filesystem/artifact_stat.txt
871eb2e5c6b26ecc...  ./filesystem/IR1-2026-002_artifacts.tar
6479cd694d0638b0...  ./markers.csv
d239add40fc9c1e3...  ./memory/syncd_core_strings.txt
3967907c437e3781...  ./network/hypervisor_nic0_clinical.pcap
f1d996df10dc4c9a...  ./network/hypervisor_nic1_quarantine.pcap
0e6c95cd676f4c91...  ./network/ioc_posts_nic0.tsv
ef0c2bf5c2d31d93...  ./network/IR1-2026-002_live_tcpdump.pcap
f1083e33fd23695b...  ./transcripts/01_triage.txt
c7e3a1dad48d9a45...  ./transcripts/02_c1_snapshot.txt
76582d80cc5fda26...  ./transcripts/03_c2_volatile.txt
6c13158383403446...  ./transcripts/04_c3_block.txt
8aeac41a6fbf65bd...  ./transcripts/05_c4_memory.txt
35333dffb9c8310b...  ./transcripts/06_c5_filesystem.txt
4a44a27e02162ca3...  ./transcripts/07_c6_isolate.txt
2e5f7c64507b03da...  ./transcripts/08_c7_shutdown.txt
68bae09961da2021...  ./transcripts/09_c8_verification.txt
76e8228dcc1f0db8...  ./volatile/network_connections.txt
3240ba6818dddec0...  ./volatile/persistence_services.txt
af5b79881da60707...  ./volatile/processes.txt
97f708cf16b9dd86...  ./volatile/suspect_process.txt
8f878522b9efea60...  ./volatile/system_state.txt
# outside git: /Users/Adult/ir1-vm-lab/evidence/IR1-2026-002
7010d346a57e2818...  IR1-2026-002_parrot-vm.mem.elf
1d4ccb577d0dcdd5...  IR1-2026-002_syncd.core
```